

Human-in-the-Loop Active Learning for Reducing Annotation Cost in Medical Image Classification

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Abstract—It is very expensive to get reliable medical-image labels, as those typically will need to be reviewed by experts. Rather than annotating each of the images, this study explores the possibility of a smaller subset of images that is sufficient to compete in chest X-ray classification with an active-learning system. The numbers of the training images and independent test images after duplicate removal were 5,206 and 624, respectively. The performance of the original CNN, DenseNet-121 and ResNet-50 models were first compared with a fixed validation split. To select the best feature extractor to be used in the active-learning stage, ResNet-50 was used for validation with the highest balanced accuracy of 0.9697 was chosen.

Three acquisition strategies compared against a stratified 5% labeled pool of 221 images at labeling budgets varying from 5% to 40% are: Random sampling, Entropy sampling, and a Hybrid uncertainty–diversity method. For each of three stochastic runs, Entropy and Hybrid achieved or matched the 100%-label reference F1-score of 0.9097 with approximately 10% of the training pool; Random first surpassed the reference around 30%. However, although Hybrid had the highest normalized area under the mean-F1 learning curve, 0.9137, the difference was also small compared to Entropy. From these result, it is found that model guided sample selection can help to minimize the number of annotations from this task. The estimated saving, however, relies on the dataset, cold start pool and fixed feature representation.

Index Terms—Active Learning, Human-in-the-Loop, Chest X-ray, Medical Image Classification, Annotation Efficiency

I. INTRODUCTION

Deep-learning models in medical imaging are often limited less by image availability than by the availability of trustworthy labels. Chest X-rays are easy to collect in large numbers, but assigning reliable diagnostic labels usually requires clinical expertise. Active learning (AL) offers a practical way to reduce that burden: instead of asking an expert to label every image, the model repeatedly identifies the unlabeled cases that appear most useful, receives labels for those cases, and learns from the enlarged labeled set [1].

The question examined here is therefore simple but important: how much of the training pool actually needs to be labeled before the classifier reaches useful performance? The project focuses on binary NORMAL versus PNEUMONIA classification. A custom CNN, DenseNet-121, and ResNet-50 are first compared on the same validation split. The validation-selected ResNet-50 is then frozen and used to obtain feature embeddings for the AL experiment. Random, Entropy, and

Hybrid uncertainty–diversity acquisition are evaluated at the same annotation budgets, with the same data split and classifier protocol. F1, balanced accuracy, ROC-AUC, AUPRC, and annotation saving are reported across three stochastic runs.

By keeping most of the pipeline fixed and changing only the acquisition rule, the experiment aims to make the effect of sample selection easier to interpret.

II. LITERATURE REVIEW

Five studies were selected because each one connects closely to a design choice in the project: expert-in-the-loop learning, uncertainty sampling, feature-space diversity, annotation reduction, class imbalance, or cold-start selection.

A. Human-in-the-Loop Medical Imaging

Budd *et al.* summarized the active learning and human-in-the-loop deep learning for medical-image analysis and discussed the role of expert involvement in data selection, model refinement, and deployment [1]. Their discussion is pertinent to this project as the model determines which cases are to be labeled, and the actual label is still provided by an annotator. The process is simulated in the present experiment. The acquisition function is not allowed to see the ground-truth label and the current label is only revealed once an image is selected. This is a simulated approximation of the decision flow found in an actual annotation environment.

B. Uncertainty and Representation-Aware Chest X-Ray Selection

For chest X-rays, Nguyen *et al.* proposed GOAL, which fuses uncertainty and representation-aware selection to save the effort of annotation on numerous near-duplicate examples [2]. This concept was directly used in this project as a Hybrid strategy. The implementation is deliberately made simpler. Entropy is applied to determine uncertain candidates, ResNet-50 embeddings are frozen and used to describe their positions in the feature space, and MiniBatch K-means is used to select representative cases. The query batch that is created is then supposed to be difficult and diverse.

C. Annotation Reduction in Radiology

Bangert *et al.* evaluated the ability of radiology models to perform as well as full-data training when trained with

a small subset of annotated data. They reported significant savings using active learning [3]. They have developed their work to facilitate the use of learning curves rather than a single final accuracy value. This strategy is followed in the present experiment, where the performance is measured at different budgets of annotations and the point at which each strategy first achieves the same-protocol full-label reference is determined. For the sake of comparison and consistency, the feature representation, the family of classifiers, the data split and the schedule of budgets are kept constant for the three acquisition methods.

D. Active Learning Under Class Imbalance

Gabriel *et al.* investigated the active learning on chest X-ray data sets with imbalanced classes and emphasized the need for class-aware training and evaluation [4]. That is also the case in our training pool, where there are significantly more pneumonia images than normal images. However, the class weights are balanced again, using the subset of classes that are currently labeled. Balanced accuracy is also reported along with the F1 and ranking-based measures.

E. Cold-Start and Hybrid Sample Selection

Yuan *et al.* compared uncertainty, diversity and hybrid cold-start strategies for chest-radiograph classification, demonstrating that the acquisition quality is highly dependent on the feature representation to be selected for the cold-start [5]. They found that the choice of one fixed ImageNet-pretrained ResNet-50 representation for all AL strategies was affected by their results. The embedding space is the same for all the three methods, which eliminates one source of variation and simplifies the comparison of the three sampling methods: Random sampling, Entropy sampling and Hybrid sampling.

In general, these studies indicate that by using active learning, labeling effort can be reduced, but the benefits can vary depending on the quality of the representation, the uncertainty estimation, the class balance and the initial labeled set. This ongoing project is narrower in scope and involves comparing three strategies of acquisition with the same split, representation, label budgets, and evaluation protocol.

III. EXPERIMENTAL DATA PROFILE

After cleaning, the development data were split into a 4,425-image training pool and a 781-image validation set, while the original 624-image test set was left untouched. Pneumonia accounts for roughly 74% of the training and validation partitions but 62.5% of the test set. This difference makes balanced metrics useful when interpreting performance. Figure 1 summarizes both the class proportions and the way the labeled pool grows as the annotation budget increases.

IV. METHODOLOGY

A. Data Preparation and Baseline Selection

26 images was removed using Duplicate screening from the original 5,232-images was in the training pool, after this 5,206 images were left for training the model. There

TABLE I
FIXED EXPERIMENTAL CONFIGURATION

Component	Setting
Task	NORMAL vs. PNEUMONIA
Input resolution	$224 \times 224 \times 3$
Training pool	4,425 images
Validation / test	781 / 624 images
Initial labeled pool	221 images ($\approx 5\%$)
Budgets	5, 10, 15, 20, 30, 40%
Feature backbone	Frozen ResNet-50
Embedding dimension	2048
Acquisition	Random, Entropy, Hybrid
Repeated seeds	42, 123, 2026
Optimizer	Adam
Batch size	32

were no overlapping between the cleaned training data and the independent test partition. Every image was resized to 224×224 pixels. Training augmentation was kept little: small rotations, embedding, zoom and contrast changes were allowed, while changing the horizontal rotation was completely avoided because right-left X-ray information matters a lot in chest radiographs. Adam optimization, early stopping, best-checkpoint saving and rollback and batch size of 32 were mostly used during the supervised baseline model training.

The class distribution of the dataset was uneven, so class weights were recomputed from the samples that were labeled at a given stage. For class c , the balanced weight and weighted sparse cross-entropy are:

$$w_c = \frac{N}{CN_c}, \quad \mathcal{L}(\theta) = -\frac{1}{m} \sum_{i=1}^m w_{y_i} \log p_{i,y_i}, \quad (1)$$

where N_c is the number of labeled samples in class c , $C = 2$, and p_{i,y_i} is the predicted probability of the true class.

Three supervised models were trained and validation and balance were compared before going through active learning. The custom CNN implemented three convolutional blocks and a global average pooling also a 128-unit dense representation. DenseNet-121 and ResNet-50 used ImageNet-pretrained convolutional removing its backbones with the original classification heads. Both transfer-learning backbones were kept frozen, then followed by global average pooling, dropouts and a two-class softmax layer. Based on validation performance only the most appropriate model was chosen. ResNet-50 achieved the highest validation balanced accuracy and was chosen as the backbone before any test-set from the data comparison was considered. Table I lists the settings which was implemented in all acquisition strategies.

B. Frozen Representation and Pool-Based Active Learning

Once ResNet-50 had been chosen as the active model usage, its weights were kept constant. The global-average-pooling output was implemented as the representation of each image,

$$\mathbf{z}_i = \text{GAP}(f_\phi(x_i)) \in \mathbb{R}^{2048}, \quad (2)$$

here f_ϕ marks the pretrained ResNet-50 feature extractor. These embeddings were cached only one time. Because of this,



Fig. 1. Pre-methodology data profile. Left: normalized class prevalence across train, validation, and test partitions. Right: labeled and remaining-unlabeled pool sizes at the tested annotation budgets.

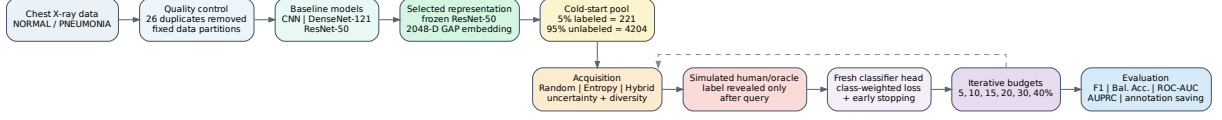


Fig. 2. Graphical abstract of the implemented pipeline: data cleaning and baseline comparison, frozen ResNet-50 feature extraction, 5% cold start, Random/Entropy/Hybrid acquisition, simulated expert label reveal, classifier retraining, and repeated evaluation at increasing label budgets.

in each cycle the algorithm only needed to retrain a small two-class head. This resulted in reduced computation, this keeps the representation similar for Random, Entropy, and Hybrid acquisition.

The training pool contained $N = 4425$ images. A stratified 5% cold-start set gives 221 labeled images, leaving 4,204 images in the initial unlabeled pool. For a budget $b \in \{0.05, 0.10, 0.15, 0.20, 0.30, 0.40\}$, the target number of labels and the number of new queries are

$$n_b = \text{round}(bN), \quad q_t = \max(0, n_b - |\mathcal{L}_t|). \quad (3)$$

Random sampling picks those q_t cases uniformly from the current unlabeled set. Where Entropy sampling sets priority of cases by predictive uncertainty,

$$H(x_i) = - \sum_{c=1}^2 p(c | x_i) \log[p(c | x_i) + \epsilon], \quad (4)$$

$$\mathcal{Q}_t = \text{Top}_{q_t} \{H(x_i) : i \in \mathcal{U}_t\}.$$

so predictions that are very close to an even class split receive greater priority.

Hybrid acquisition adds diversity after the uncertainty step. It first considers up to $3q_t$ candidates with high-entropy, then normalizing those candidates embeddings as $\tilde{\mathbf{z}}_i = \mathbf{z}_i / \|\mathbf{z}_i\|_2$ and applies MiniBatch K-means. The representative selection and clustering objectives are

$$\min_{\{\mu_k\}} \sum_{k=1}^{q_t} \sum_{i \in C_k} \|\tilde{\mathbf{z}}_i - \mu_k\|_2^2, \quad i_k^* = \arg \min_{i \in C_k} \|\tilde{\mathbf{z}}_i - \mu_k\|_2. \quad (5)$$

In simulation terms, the method first wants to find the images that are difficult for the classifier and then abstain from selecting and inputting difficult images that almost same in the learned feature space in the query batch.

C. Simulated Human-in-the-Loop and Repeated Evaluation

The “human Annotating” step was simulated rather than conducted by a radiologist. Furthermore, the acquisition functions do not receive the labels of the unlabeled pool. Only after a query set $\mathcal{Q}_t$ has been implemented then the existing ground-truth labels are revealed. This changing is written as:

$$\mathcal{O}(x_i) = y_i \ (i \in \mathcal{Q}_t), \quad \mathcal{L}_{t+1} = \mathcal{L}_t \cup \mathcal{Q}_t, \quad \mathcal{U}_{t+1} = \mathcal{U}_t \setminus \mathcal{Q}_t. \quad (6)$$

The enlarged labeled set is then used to train a fresh class-weighted classifier head. This setup does not neglect the importance of the time, disagreement or the judgment of a real radiologist, but it does reproduce the order of decisions in a pool-based HITL loop: selecting first, obtaining the label second and retraining afterward.

Repeated runs were done using seeds 42, 123 and 2026. In the current implementation the same initial 5% pool is kept constant throughout the runs while optimization and acquisition randomness may be different. For any metric M over $R = 3$ runs,

$$\bar{M} = \frac{1}{R} \sum_{r=1}^R M_r, \quad s_M = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (M_r - \bar{M})^2}. \quad (7)$$

Annotation saving is calculated from the first tested budget that is a match or exceeds with the F1-score of the same-protocol with 100%-label reference. If $M_s(b_k)$ is the donated as F1 for strategy s at budget b_k and M_{ref} is the reference value,

$$b_s^* = \min\{b_k : M_s(b_k) \geq M_{ref}\}, \quad S_s = 100 - b_s^*. \quad (8)$$

As a single crossing point is important for the chosen budget grid. Furthermore, normalized area under the learning curve

TABLE II
VALIDATION PERFORMANCE USED FOR BACKBONE SELECTION

Model	F1	Bal.Acc.	ROC-AUC	AUPRC
CNN	0.9591	0.9288	0.9772	0.9920
DenseNet-121	0.9665	0.9543	0.9908	0.9967
ResNet-50	0.9809	0.9697	0.9969	0.9989

TABLE III
INDEPENDENT TEST CHECK OF THE BASELINE MODELS

Model	Recall	F1	Bal.Acc.	ROC-AUC
CNN	0.9641	0.8806	0.7940	0.9368
DenseNet-121	0.9641	0.9447	0.9179	0.9795
ResNet-50	0.9872	0.9244	0.8697	0.9829

marked by AULC was also calculated,

$$\text{AULC}_s = \frac{1}{b_K - b_1} \sum_{k=1}^{K-1} \frac{M_s(b_k) + M_s(b_{k+1})}{2} (b_{k+1} - b_k). \quad (9)$$

This presents a wider view of labeling efficiency throughout the entire 5–40% range than evaluating a strategy just from a budget alone. F1 was highlighted because it helps to maintain the balances between precision and recall under class imbalance, while balanced accuracy, ROC-AUC, and AUPRC were implemented as supporting measures.

V. RESULTS AND DISCUSSION

A. Baseline Model Selection

Table II gives the validation results that was used in the backbone selection. ResNet-50 performed better than other and stood first on balanced accuracy (0.9697), F1 (0.9809), ROC-AUC (0.9969), and AUPRC (0.9989), so it was chosen before looking at independent-test rankings.

The independent test results in Table III are useful because they visualize that the validation ranking did not simply repeat on the test set. DenseNet-121 had the highest test results in the F1 and the balanced accuracy, while ResNet-50 had the highest recall and ROC-AUC. ResNet-50 was nevertheless retained for the Active-Learning stage. A change of the backbone after seeing test results would have made the test set part of the model-selection process.

B. Active-Learning Efficiency

The same-protocol classifier trained with 100% of the cached training embeddings reached $F1 = 0.9097$, balanced accuracy = 0.8372, ROC-AUC = 0.9874, and AUPRC = 0.9923. It is to be read as the full-label reference for the embedding-plus-head AL Algorithm but not as a replacement for the previous baseline models that was supposed to be used as the selector of the feature extractor.

Table IV reports mean F1 and standard deviation across the three runs. All strategies have shown the same 5% result because they started the run from the same cold-start pool; the acquisition rules only starts to matter only when more samples are inputted in the query.

TABLE IV
ACTIVE-LEARNING F1 ACROSS THREE RUNS (MEAN $\pm$ SD)

Labels (%)	Random	Entropy	Hybrid
5	0.8954 $\pm$ 0.0140	0.8954 $\pm$ 0.0140	0.8954 $\pm$ 0.0140
10	0.9096 $\pm$ 0.0024	0.9151 $\pm$ 0.0101	0.9161 $\pm$ 0.0065
15	0.9096 $\pm$ 0.0203	0.9135 $\pm$ 0.0075	0.9141 $\pm$ 0.0171
20	0.8997 $\pm$ 0.0067	0.9103 $\pm$ 0.0062	0.9156 $\pm$ 0.0154
30	0.9138 $\pm$ 0.0041	0.9173 $\pm$ 0.0084	0.9179 $\pm$ 0.0126
40	0.9114 $\pm$ 0.0049	0.9109 $\pm$ 0.0048	0.9085 $\pm$ 0.0063

TABLE V
ANNOTATION-EFFICIENCY SUMMARY

Strategy	F1@10%	Match Budget	Saving	F1 AULC
Random	0.9096	30.01%	69.99%	0.9080
Entropy	0.9151	9.99%	90.01%	0.9125
Hybrid	0.9161	9.99%	90.01%	0.9137

At roughly 10% labels, mean F1 was 0.9096 ± 0.0024 for Random, 0.9151 ± 0.0101 for Entropy, and 0.9161 ± 0.0065 for Hybrid. Entropy and Hybrid therefore reached the full-label reference at this budget, while Random could not even reach it until about 30%. Hybrid showed its highest mean F1 at 30% (0.9179 ± 0.0126). The balanced-accuracy curve represents a similar story from a class-sensitive perspective and avoids an apparent gain caused only by the majority class.

Table V summarizes the first tested budget that reaches the reference and the normalized AULC over 5–40%. Hybrid showed the largest F1 AULC, but Entropy almost the same. The difference interprets as a modest numerical advantage than as evidence that Hybrid is universally superior.

C. What the Acquisition Strategies Are Doing

Random sampling is the necessary control: it shows what happens when the annotation budget is kept same without using the model to guide selection. Entropy tries to spend that budget on cases close to the current decision boundary, where the classifier is least certain. Then Hybrid strategy begins from the same uncertainty idea but then makes sure whether the candidates that were selected are spread across the 2048-dimensional ResNet-50 feature space. This extra step is implemented to make sure to prevent a query batch from containing several same difficulty level examples that are identical.

The results support the value of informative acquisition more clearly than they prove that Hybrid has a large advantage over Entropy. Hybrid had the highest AULC (0.9137), but Entropy follows closely at 0.9125 and the ranking changed at later budgets. That behavior is plausible because AL performance depends mostly on the quality of confidence calibration, the feature representation and the composition of the current unlabeled pool [2], [5]. The non-monotonic curves are not unexpected: samples that were newly added has a possibility to change class balance and boundary difficulty and the classifier head is always reinitialized for each budget.

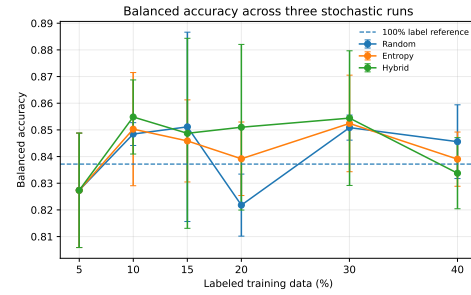
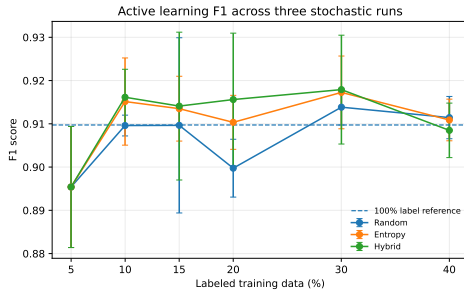


Fig. 3. Three-run active-learning performance. Left: mean F1 with standard-deviation error bars. Right: mean balanced accuracy. Dashed horizontal lines indicate the corresponding same-protocol 100%-label reference values.

D. How to Read the Annotation-Saving Result

The reported 90% saving for Entropy and Hybrid is a saving in *number of labels requested*, not a measured 90% reduction in radiologist time or financial cost. The calculation considers all the annotations as having the same valuation. Human annotation is more complicated: difficult images are sure to take longer to interpret, some cases require intensive study and difference in interpretation can lead to additional review. Even with this limitation, the project shows a useful principle. Under a common representation and classifier protocol, the model-guided selection showed the chosen performance target after labeling a much smaller portion of the of the unlabeled pool, which is consistent with earlier radiology Active Learning works [3].

E. Relation to Earlier Work

The low-budget results are showing similar results with medical-image AL studies that report near-full-data performance from substantially smaller labeled subsets [2], [3]. What this experiment does is that it adds a controlled internal comparison: Random, Entropy and Hybrid using the same ResNet-50 representation and the similar budget schedule. Within that setting, Random needed more labels to reach the desired F1, while Entropy and Hybrid did it much earlier. The small AULC gap between the two informed strategies is also useful. It shows that implementing diversity can improve the result but it never guarantees a huge difference in result when the uncertainty results and underlying representation are already high [5].

F. Clinical Translation and Deployment Prospect

The pipeline could operate as an annotation-priority tool rather than as an automatic diagnostic system. A hospital could start with a small expert-labeled seed set then extract features for the remaining archive, train a classifier, and consider to send only selected uncertain or diverse studies to a human radiologist. After those labels are inputted, the training set would be bigger and the model could work with the be updated pool. The acquisition function then can decide *which images deserve expert attention first*; it does not mean to replace the clinician ground-truth diagnosis [1].

This idea is not only limited to normal vs pneumonia classification. With appropriate training and functions, it can be

made to classify multi-label chest radiography, CT scans, MRI, dermatology, histopathology or segmentation. Another natural direction is to combine AL with semi-supervised learning, so only some queried images are needed expert labeling while sure predictions with confident of much larger unlabeled set still contribute to training. In a deployment setting, active learning may also be in use to identify distribution changes, needing new labeling only when the selected cases differ from the dataset which was used to train initially.

A more industrial level system should also model annotation cost directly. Some images can be labeled quickly, whereas more complicated cases may require comparison with previous studies or a second expert opinion. Recording annotation time, confidence and a difference in opinion would allow a future acquisition rule to optimize information gained per unit of expert effort instead of evaluating all image cost the same .

G. Threats to Validity

Some design choices make the comparison internally consistent: the validation and test partitions are fixed, all methods use the same label budgets and frozen feature space and the same evaluation protocol is applied to every strategy. At the same time, the project does not capture all the source of uncertainty. Same cold-start pool is shared in the three runs , the labels come from a simulated method and the backbone is always kept fixed rather than fine-tuned every time. External validity is also limited by the use of one dataset and one binary classification task. Finally, annotation percentage is a dummy setup for real expert workload. The findings should therefore be read as a controlled proof-of-concept not as a universal estimate of clinical savings.

VI. LIMITATIONS AND FUTURE WORK

The main limitation of this experiment is that no radiologist participated in the annotation process and also only a single chest X-ray dataset trained the models. Moreover, the ResNet-50 backbone and fine-tuning remained frozen and only three stochastic runs were performed. Because these cycle shared the same initial 5% labeled pool, the resulted standard deviations capture training and acquisition randomness but not sensitivity to a different cold start. Softmax entropy can also be overconfident mostly when the data distribution is varying.

A stronger future experiment would repeat the study with 5–10 sampled cold-start pools that are not connected with each-other, repeating the 100%-label reference under the same seeds, and adding more finer budgets between 5% and 15% to locate the first reference crossing more accurately. Other useful segments can include calibrated or Bayesian uncertainty, a pure-diversity baseline, partial or end-to-end backbone fine-tuning, external test sets from another institution, multiple-label disease prediction and a real human annotation system that evaluates expert time and disagreement.

VII. CONCLUSION

This project implemented a reproducible simulated human-in-the-loop AL pipeline for binary chest X-ray classification. ResNet-50 was selected through validation balanced accuracy and then kept frozen so that Random, Entropy and Hybrid acquisition could be compared in the same 2048-dimensional feature space. Across the tested 5–40% labeling budgets, Entropy and Hybrid reached the same-protocol full-label reference F1 using only 10% of the training pool, while Random first exceeded the reference around 30%. Hybrid produced the largest mean-F1 AULC, although its margin over Entropy was small and was not tested for statistical significance.

The important point is not that less data is automatically better. Rather, the project shows that *which examples are labeled* can matter almost as much as how many are labeled. Under the controlled conditions used here, uncertainty-aware selection, with or without an added diversity step, concentrated annotation effort on more informative cases and reached competitive performance with a fraction of the available training labels.

VIII. REPRODUCIBILITY AND DATA AVAILABILITY

Fixed train, validation, and test manifests were stored together with active-learning pool indices, cached embeddings, model checkpoints, and per-seed result files. The chest X-ray dataset used in the project is publicly available from its original source. Keeping these intermediate artifacts makes it possible to reproduce the budget calculations and acquisition comparisons without redefining the data split.

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